

Matias Freire

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EDUCATION

Stevens Institute Of Technology

Expected May 2028

B.S. in Computer Science

Hoboken, NJ

Coursework: Algorithms and Data Structures, Computer Architecture, Systems Programming, Software Development Processes

Certifications: Bloomberg Market Concepts, Hugging Face AI Agents Fundamentals

Cornell Tech – Break Through Tech

Summer 2026

Machine Learning Fellow

Remote

- Building advanced skills in machine learning, deep learning, and neural networks through an intensive Cornell Tech program, culminating in the AI Studio Fellowship , a paid team-based placement partnering with industry sponsors to develop real-world ML solutions.

EXPERIENCE

Software Engineering Intern

May 2026 – Present

Wabtec Corporation

Erie, PA

- Developing a Raspberry Pi-based intelligent panel to visualize real-time train data by integrating Data Distribution Service (DDS) communication with embedded system workflows.
- Encapsulating train data monitoring functionality into a user-facing application by building a custom UI with Python, C, Tailwind CSS, and HTML.

Deep Learning Research Assistant

Aug 2025 – Mar 2026

Advanced Quantum Materials Lab

Hoboken, NJ

- Developed a Dash-Plotly platform integrating literature-driven preprocessing, synthetic data, and open-source pipelines for interactive MoS defect visualization and DL training.
- Studied preprocessing methods, synthetic data generation, and deep learning models (CNNs like MobileNet), Extraordinary Magnetoresistance (EMR) devices and learned how to replicate it's magnoreistence results on COMSOL.

PROJECTS

SHPE and Colorstack Stevens Chapter Website | *Next.js, React, Tailwind, Typescript*, Supabase [Github-Repo](#)

- Built from scratch modular, responsive websites for the SHPE and ColorStack Stevens chapters, **serving 50+ and 150+ members as central platforms** to attract corporate sponsors and onboard new members
- Implemented reusable UI components in TypeScript and Tailwind CSS, integrating **authentication and CRUD** functionality, and **leveraged the Supabase MCP within Google's Antigravity environment** to accelerate full-stack development

MoS2 Image Synthesis & Analysis Platform | *PyTorch, Dash-Plotly, CSS, Docker*

[Github-Repo](#)

- Designed and built from scratch an end-to-end Dash-Plotly research platform for deep learning-based MoS defect analysis
- Automated synthetic image generation with configurable defects and realistic noise injection, integrating a ResUNet model for vacancy and polymorph classification

LEADERSHIP

Vice-President - Previous Head of External Relations | *Colorstack Stevens Chapter*

April 2025 – Present

- Led growth and technical engagement initiatives, **increasing membership by 150% and online engagement by 125% through social media** strategy and **15+ events per semester**
- Serve as the primary representative for partnerships and collaborations with major campus organizations, including CS Club, SHPE, NSBE, and others
- Organized peer-led weekly technical sessions, guiding members through 32+ Leetcode Questions

Chief Web Officer | *SHPE Stevens Chapter*

March 2026 – Present

- Leading development of the official chapter website in Next.js/Supabase, building the platform from scratch to **support 150+ registered members** and present the organization to corporate sponsors
- lanning and organizing technology-focused events and workshops to expand technical engagement across the chapter

STEP Tutor-Counselor/Resident Assistant | *STEP Bridge Summer Program*

Jun 2025 – Aug 2025

- Led Calculus I/II and Physics recitations for **52+ incoming freshmen**, strengthening STEM readiness through structured reviews and weekly after-hours **tutoring for 25+ student**
- Served as a **Resident Assistant**, **mentoring 52+ residents** while enforcing residential policies and supporting academic success, wellness, and community development

TECHNICAL SKILLS

Programming: Java, Python, C/C++, JavaScript

Web Tools: TypeScript, React, Tailwind, Next.js, Supabase

Developer Tools: Git, GitHub, Docker , Jupyter Notebooks, Linux, Dash

Libraries: Dash-Plotly, Pandas, NumPy, PyTorch, Scikit-learn, Streamlit